

$$\phi = 3.1 \times 10^{-19} \text{ J}$$

29

B

From $\frac{hc}{\lambda} = \phi + KE_{\text{max}}$:

As the intensity of light affects the number of photons incident on the metal rather than the energy carried by each photon, $\frac{hc}{\lambda}$, the maximum KE of an electron is independent of the intensity.

30

A

The diagram shows that most of the alpha particles passed through the gold foil